Equality and Stability in the Entropy Power Inequality: Extensions to Random Vectors and Log-Concave Sums

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Research internship report
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Abstract

Shannon's entropy power inequality (EPI) states that, for independent random variables X, Y in $\mathbb{R}$ and $\lambda \in (0,1)$,

$$h\left(\sqrt{\lambda}X + \sqrt{1-\lambda}Y\right) \geq \lambda h(X) + (1-\lambda)h(Y),$$

with equality if and only if X and Y are Gaussian with the same variance.

Prior proofs of the EPI had only settled the equality case under additional restrictions, such as finite second moments or regularity assumptions on the densities of X and Y, due to the discontinuity phenomenon exhibited by the counterexample of Bobkov and Chistyakov.

In the multivariate setting, while the equality case of the EPI in $\mathbb{R}^d$ has long been known, with equality if and only if X and Y are Gaussian with proportional covariance matrices, existing proofs continue to rely on second moment or regularity assumptions. Building on the recent work of Gavalakis and Kontoyiannis, who characterized the equality case and established a stability result for the EPI under minimal assumptions in dimension one, this report extends these results to random vectors in $\mathbb{R}^d$.

We first show that the pathological behaviour exhibited by the Bobkov-Chistyakov counterexample is the only obstruction to the continuity of $t \mapsto h(X + \sqrt{t}Z)$ at $t = 0$ in $\mathbb{R}^d$. We then establish de Bruijn's identity in this same general setting, without any moment assumptions:

$$\frac{d}{dt}h\left(X + \sqrt{t}Z\right) = \frac{1}{2}I\left(X + \sqrt{t}Z\right), \qquad t > 0.$$

These two results allow us to characterize the equality case in Shannon's EPI in $\mathbb{R}^d$ and to establish a qualitative stability result under the Lévy-Prokhorov metric.

We further obtain a discrete counterpart of the EPI for sums of independent, not necessarily identically distributed, log-concave random variables on $\mathbb{Z}$. Under this assumption, we establish an approximate entropy power inequality together with an associated stability result:

$$H\left(\sum_i X_i\right) \geq \frac{1}{n}\sum_i H(X_i) + \frac{1}{2}\log n - o(1).$$

This report presents the context of the research internship during which these results were obtained, carried out at the University of Cambridge in the Statistical Laboratory within the Department of Pure Mathematics and Mathematical Statistics. This internship was done during the first year of master's degree (M1) in mathematics at ENS Paris-Saclay.

It also provides a review of the relevant state of the art, together with an introduction to the basic notions of information theory intended for readers unfamiliar with the field.

Mathematics Subject Classification: 94A17, 60E15, 62B10, 60F05, 60E05, 52A40.

Résumé

L'inégalité de puissance entropique (EPI) de Shannon énonce que, pour des variables aléatoires indépendantes X, Y dans $\mathbb{R}$ et $\lambda \in (0,1)$,

$$h\left(\sqrt{\lambda}X + \sqrt{1-\lambda}Y\right) \geq \lambda h(X) + (1-\lambda)h(Y),$$

avec égalité si et seulement si X et Y sont gaussiennes de même variance.

Les preuves antérieures de l'EPI n'établissaient le cas d'égalité que sous des hypothèses supplémentaires, telles que le second moment fini ou des hypothèses de régularité sur les densités de X et Y, en raison du phénomène de discontinuité mis en évidence par le contre-exemple de Bobkov et Chistyakov.

En dimension $d > 1$, si le cas d'égalité de l'EPI dans $\mathbb{R}^d$ est connu de longue date, avec égalité si et seulement si X et Y sont gaussiennes à matrices de covariance proportionnelles, les preuves existantes continuent de reposer sur des hypothèses de second moment ou de régularité. S'appuyant sur les travaux récents de Gavalakis et Kontoyiannis, qui ont caractérisé le cas d'égalité et établi un résultat de stabilité pour l'EPI sous hypothèses minimales en dimension un, ce rapport étend ces résultats aux vecteurs aléatoires dans $\mathbb{R}^d$.

Nous montrons tout d'abord que le comportement pathologique exhibé par le contre-exemple de Bobkov-Chistyakov constitue la seule obstruction à la continuité de $t \mapsto h(X + \sqrt{t}Z)$ en $t = 0$ dans $\mathbb{R}^d$. Nous établissons ensuite l'identité de de Bruijn dans ce même cadre général, sans aucune hypothèse de moment :

$$\frac{d}{dt}h\left(X + \sqrt{t}Z\right) = \frac{1}{2}I\left(X + \sqrt{t}Z\right), \qquad t > 0.$$

Ces deux résultats nous permettent de caractériser le cas d'égalité dans l'EPI de Shannon dans $\mathbb{R}^d$ et d'établir un résultat de stabilité qualitative sous la distance de Lévy-Prokhorov.

Nous obtenons de plus un pendant discret de l'EPI pour des sommes de variables aléatoires indépendantes, non nécessairement identiquement distribuées, log-concaves sur $\mathbb{Z}$. Sous cette hypothèse, nous établissons une inégalité de puissance entropique approchée ainsi qu'un résultat de stabilité associé :

$$H\left(\sum_i X_i\right) \geq \frac{1}{n}\sum_i H(X_i) + \frac{1}{2}\log n - o(1).$$

Ce rapport présente le contexte du stage de recherche durant lequel ces résultats ont été obtenus, effectué à l'Université de Cambridge au sein du Statistical Laboratory, dans le Department of Pure Mathematics and Mathematical Statistics. Ce stage a été réalisé au cours de la première année de master (M1) de mathématiques à l'ENS Paris-Saclay.

Il propose également une revue de l'état de l'art pertinent, ainsi qu'une introduction aux notions de base de la théorie de l'information à destination des lecteurs non familiers avec le domaine.

Mathematics Subject Classification: 94A17, 60E15, 62B10, 60F05, 60E05, 52A40.

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Large Language Models (LLMs) were used to assist with the Latex formatting and to improve the syntactic formulation of certain sentences in this report. All mathematical results and proofs are my own original work.

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1 Introduction

Shannon entropy has occupied a central place in information theory since its introduction in 1948: it quantifies the uncertainty associated with a random variable, and allows to measure the information communicated through a channel. Its continuous counterpart, differential entropy, plays an equally fundamental role, in particular through one of its most useful properties: the entropy power inequality (EPI), first stated by Shannon himself and rigorously proved by Stam in 1959. This inequality asserts that, for two independent random variables, the entropy of their scaled sum is bounded below by their individual entropies, with equality if and only if both variables are Gaussian with the same variance. Beyond its intrinsic interest, the EPI is used in the proof of various results including the computation of capacity bounds for gaussian channels.

While the inequality itself is well understood, characterizing its equality case under minimal assumptions remained an open problem for a long time. All the historical proofs rely on an additional integrability assumption (a finite second moment, or regularity of the densities) that is not required for the statement of the inequality itself. This assumption is necessary because of a discontinuity phenomenon exhibited by the Bobkov-Chistyakov [4] counterexample. It is only recently that Gavalakis and Kontoyiannis showed that this pathological behaviour is in fact the only obstruction, obtaining the first characterization of the equality case of the EPI on the real line under the only assumption that the relevant entropies are finite.

This report presents the work carried out during a four-month research internship, undertaken at the Statistical Laboratory of the Department of Pure Mathematics and Mathematical Statistics of the University of Cambridge, under the supervision of Prof. I. Kontoyiannis, as part of the first year (M1) of the Hadamard master's program at ENS Paris-Saclay. The goal of this internship was to extend the results of Gavalakis and Kontoyiannis in two directions: first, by extending their characterization of the equality case and their stability result from the real line to random vectors in $\mathbb{R}^d$ and second, by establishing a discrete counterpart of these results for sums of independent log-concave random variables taking values in $\mathbb{Z}$.

The report is organized as follows. Section 2 gathers the information-theoretic prerequisites needed to follow the report.

Section 3 provides a review of the relevant state of the art. It traces the development of the EPI, from the foundational work of Shannon and Stam to the most recent results of Gavalakis and Kontoyiannis.

Section 4 contains the first contribution of the report: the extension to the multivariate setting of the results of Gavalakis and Kontoyiannis.

Section 5 develops the second contribution of the report, in the discrete setting on $\mathbb{Z}$. We establish a quantitative approximation of the discrete entropy of a sum of log-concave random variables by smoothing with independent uniform random variables, which yields an approximate entropy power inequality for sums of independent log-concave random variables, together with an associated stability result.

Section 6 discusses the limitations of the approach taken and possible directions for extending the discrete results.

Finally, Section 7 concludes the report, summarizing the scientific contributions obtained and reflecting on the personal and academic experience that this research internship represented.

All the proofs are in the appendix A, so as not to overload the main body of the report.

2 Information Theory Prerequisites

This section introduces the basic notions and results in information theory needed to follow the contributions presented in this report, intended for readers who are not already familiar with the field.

2.1 Shannon Entropy and Differential Entropy

Entropy quantifies the amount of uncertainty, associated with the outcome of a random variable: the more spread out or unpredictable the distribution, the higher its entropy. It was first introduced by Claude Shannon in 1948 to give a mathematical formalism to data communication.

Definition 2.1. Let X be a random variable with values in a discrete set $\mathcal{X}$, and let p denote its probability mass function on $\mathcal{X}$. The (Shannon) entropy of X is defined as

$$H(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x) = -\mathbb{E}[\log p(X)], \tag{1}$$

with the convention $0 \log 0 := 0$. Throughout this report, logarithms are natural logarithms.

Definition 2.2. Let X be a random vector in $\mathbb{R}^d$ admitting a density f with respect to the Lebesgue measure. The differential entropy of X is defined as

$$h(X) := -\int_{\mathbb{R}^d} f(x) \log f(x)\, dx = -\mathbb{E}[\log f(X)], \tag{2}$$

whenever this integral exists. We adopt the convention that $h(X) = -\infty$ whenever the integral does not exist, including the case where X does not admit a density with respect to the Lebesgue measure. We say that $h(X)$ is finite when $-\infty < h(X) < \infty$.

Remark 2.3. Note that when X is instead absolutely continuous with respect to the counting measure on a discrete set $\mathcal{X}$, the integral above reduces to a sum over $\mathcal{X}$, and we recover the discrete entropy $H(X)$ defined above.

We now introduce a way to see how different one probability measure is to another with the Kullback-Leibler divergence.

Definition 2.4. Let P and Q be two probability distributions on the same space. If P has density (or probability mass function) p and Q has density (or probability mass function) q, and P is absolutely continuous with respect to Q, the Kullback-Leibler divergence (or relative entropy) between P and Q is defined as

$$D(P\|Q) := \int p(x) \log \frac{p(x)}{q(x)}\, dx = \mathbb{E}_P\left[\log \frac{p(X)}{q(X)}\right], \tag{3}$$

with the convention $0 \log \frac{0}{q(x)} := 0$. If P is not absolutely continuous with respect to Q, we set $D(P\|Q) := +\infty$. In the discrete case, the integral is replaced by a sum over the support of p. By Jensen's inequality, $D(P\|Q) \geq 0$, with equality if and only if $P = Q$.

2.2 The Entropy Power Inequality (EPI)

Having defined differential entropy, we now give the statement to one of its most famous properties: the Entropy Power Inequality (EPI), which relates the entropy of a convolution of independent random variables to the entropies of those random variables.

Theorem 2.5 (Shannon's Entropy Power Inequality [8, 3])**.** *Let X, Y be independent random variables in $\mathbb{R}$ with finite differential entropies. Then, for any $\lambda \in (0,1)$,*

$$h\left(\sqrt{\lambda}X + \sqrt{1-\lambda}Y\right) \geq \lambda h(X) + (1-\lambda)h(Y). \tag{4}$$

Equivalently, writing $N(X) := \frac{1}{2\pi e} e^{2h(X)}$ for the entropy power of X,

$$N(X+Y) \geq N(X) + N(Y). \tag{5}$$

Remark 2.6. The EPI was first stated by Shannon in his 1948 paper [8], but without a complete proof. The first rigorous proof was given by Stam in 1959 [3], using de Bruijn's identity, which we introduce in the next section. Equality holds in the EPI if and only if X and Y are Gaussian with the same variance. We return to this equality case, and to its characterization under minimal assumptions, in Section 4.

2.3 De Bruijn's Identity

To prove rigorously the EPI, a crucial tool is De Bruijn's identity, which describes the behavior of the entropy of a random variable perturbed with independent gaussian noise.

Definition 2.7. Let X be a random variable in $\mathbb{R}$ admitting an absolutely continuous density f. The Fisher information of X is defined as

$$I(X) := \int_{\mathbb{R}} \frac{f'(x)^2}{f(x)}\, dx. \tag{6}$$

If X does not admit an absolutely continuous density, we set $I(X) := \infty$.

Theorem 2.8 (De Bruijn's Identity [5])**.** *Let X be a random variable in $\mathbb{R}$ with finite differential entropy and finite variance, and let $Z \sim \mathcal{N}(0,1)$ be independent of X. Then, for every $t > 0$,*

$$\frac{d}{dt} h\left(X + \sqrt{t}Z\right) = \frac{1}{2} I\left(X + \sqrt{t}Z\right). \tag{7}$$

Remark 2.9. Writing g_t for the density of $X + \sqrt{t}Z$, one can check that g_t satisfies the heat equation $\partial_t g_t = \frac{1}{2}\Delta g_t$. De Bruijn's identity can thus be interpreted as an entropic counterpart of the heat equation.

2.4 Lévy-Prokhorov Distance and Weak Convergence

In this section, we give some prerequisites that will be used in the section 4.4 for the qualitative stability related to the continuous EPI. The proofs of these classical results can be found in Billingsley's textbook [11].

Definition 2.10. A sequence $(\mu_n)_{n\geq 1}$ of probability measures on $\mathbb{R}^d$ is said to be tight if, for every $\varepsilon > 0$, there exists a compact set $K \subset \mathbb{R}^d$ such that

$$\mu_n(K) \geq 1 - \varepsilon, \qquad \text{for all } n \geq 1. \tag{8}$$

When X_n is a random vector with law μ_n, we also say that the sequence $(X_n)_{n\geq 1}$ is tight.

Definition 2.11. Let μ, ν be two probability measures on $\mathbb{R}^d$, equipped with its Borel σ-algebra. The Lévy-Prokhorov distance between μ and ν is defined as

$$d_{LP}(\mu,\nu) := \inf\left\{\varepsilon > 0 \,:\, \mu(A) \leq \nu(A^\varepsilon) + \varepsilon \text{ and } \nu(A) \leq \mu(A^\varepsilon) + \varepsilon, \text{ for all Borel sets } A\right\}, \tag{9}$$

where $A^\varepsilon := \{x \in \mathbb{R}^d : \inf_{y\in A} \|x - y\| < \varepsilon\}$ denotes the open ε-neighborhood of A. When X, Y are random vectors with laws μ, ν respectively, we write $d_{LP}(X, Y) := d_{LP}(\mu, \nu)$.

Theorem 2.12. *The Lévy-Prokhorov distance d_{LP} metrizes weak convergence on the space of probability measures on $\mathbb{R}^d$: for a sequence $(\mu_n)_{n\geq 1}$ and a probability measure μ on $\mathbb{R}^d$,*

$$\mu_n \xrightarrow{d} \mu \quad \Longleftrightarrow \quad d_{LP}(\mu_n, \mu) \to 0, \text{ as } n \to \infty, \tag{10}$$

where $\xrightarrow{d}$ denotes convergence in distribution (weak convergence).

Theorem 2.13 (Prokhorov's Theorem)**.** *Let $(\mu_n)_{n\geq 1}$ be a tight sequence of probability measures on $\mathbb{R}^d$. Then $(\mu_n)_{n\geq 1}$ is relatively compact for the topology of weak convergence, or equivalently for the Lévy-Prokhorov distance d_{LP}: every subsequence of $(\mu_n)_{n\geq 1}$ has a further subsequence that converges weakly to some probability measure μ on $\mathbb{R}^d$.*

We now give a corollary to this theorem that will be used to assert the tightness of a sequence of probability distributions of random vectors satisfying a uniform bound on their k-th absolute moments. Tightness is often obtained via this method, and such a moment bound is precisely one of the hypotheses of our qualitative stability theorem (Theorem 4.9).

Corollary 2.14. *Let $(X_n)_{n\geq 1}$ be a sequence of random vectors in $\mathbb{R}^d$ with uniformly bounded k-th absolute moments for some $k > 1$, i.e., $\sup_n \mathbb{E}[\|X_n\|^k] < \infty$. Then the sequence of laws of $(X_n)_{n\geq 1}$ is tight, and consequently, by Theorem 2.13, every subsequence of $(X_n)_{n\geq 1}$ has a further subsequence converging weakly to some random vector X_∞ in $\mathbb{R}^d$.*

2.5 Log-Concave Random Variables and Discrete Entropy

We conclude this preliminary section with a brief overview of log-concave random variables on $\mathbb{Z}$, a fundamental family of discrete distributions with strong regularity properties.

Definition 2.15. A probability mass function $p : \mathbb{Z} \to \mathbb{R}_+$ is said to be log-concave if

$$p(k)^2 \geq p(k-1)\, p(k+1), \qquad \text{for all } k \in \mathbb{Z}. \tag{11}$$

We say that a discrete random variable X is log-concave if its probability mass function is log-concave.

Many classical discrete distributions are log-concave, including the binomial, Poisson, geometric, and hypergeometric distributions.

Theorem 2.16 (Hoggar's Theorem [7])**.** *Let $X_1, \ldots, X_n$ be independent log-concave random variables on $\mathbb{Z}$. Then $X_1 + \cdots + X_n$ is also log-concave on $\mathbb{Z}$.*

Remark 2.17. Hoggar's theorem shows that the class of discrete log-concave distributions on $\mathbb{Z}$ is closed under convolution. This property is used repeatedly in Section 5, where it guarantees that sums of independent log-concave random variables remain log-concave.

3 Related Work

In this section, we provide an overview of the state of the art and trace the historical development of the EPI.

3.1 Shannon, Stam, and Barron: Foundations of the EPI and De Bruijn's Identity

The first statement of the EPI was given by Shannon in his 1948 paper [8]. The statement was formulated in terms of the entropy power $N(X) := \frac{1}{2\pi e} e^{2h(X)}$, asserting that for independent random variables X and Y, the entropy power of their sum is bounded below by the sum of their entropy powers:

$$N(X+Y) \geq N(X) + N(Y).$$

It is a well-known fact that this entropy power formulation is perfectly equivalent to the concavity-style statement often favored in modern literature and for the proofs. Specifically, it is equivalent to asserting that for any $\lambda \in (0,1)$,

$$h\left(\sqrt{\lambda}X + \sqrt{1-\lambda}Y\right) \geq \lambda h(X) + (1-\lambda)h(Y).$$

This equivalence can be easily derived using the standard scaling properties of differential entropy (see Lieb's paper [12]). Shannon's justification for the inequality is variational and although the proof is not complete, the author gives the correct sufficient conditions for the equality case when the two variables are gaussian with proportional covariance (the equality case for the concave form is achieved for two gaussians with the same variance).

The first rigorous proof of the EPI was given by Stam [3] in 1959, who made use of de Bruijn's identity. This identity establishes a link between entropy and Fisher information (6): for a random variable X and an independent standard Gaussian $Z \sim \mathcal{N}(0,1)$,

$$\frac{d}{dt}h(X+\sqrt{t}Z) = \frac{1}{2}I(X+\sqrt{t}Z), \quad t>0.$$

In addition to de Bruijn's identity, Stam's proof relies on the Fisher information inequality. This inequality states that for independent random variables X and Y with finite Fisher information,

$$I(X+Y)^{-1} \geq I(X)^{-1} + I(Y)^{-1}. \tag{12}$$

However, the exchange of differentiation and expectation in the proof of de Bruijn's identity implicitly requires the variables to have a finite second moment, which was not mentioned in Stam's paper. Blachman [13] in 1965 clarified Stam's proof and added regularity assumptions on the densities. It wasn't until Barron [5] in 1984 that the first rigorous justification of De Bruijn's identity with a correct proof of the exchange of derivation and expectation was done under finite second moment assumption. The first qualitative stability result was done in 1991 by Carlen and Soffer [14]

Although the statement of Shannon was multidimensional, the first $\mathbb{R}^d$ proof was given in Blachman's paper [13] by proceeding with an induction on the dimension. In dimension d the entropy power is normalised per coordinate, $N(X) := \frac{1}{2\pi e}e^{2h(X)/d}$. The induction is carried out on the concave form rather than on the entropy powers themselves.

De Bruijn's identity was only established in the scalar case in [3], and its vector counterpart had to wait for Costa and Cover [15], who replaced the scalar integration by parts with an application of Green's identity to the heat equation. Once the identity is available for random vectors, and since the Fisher information inequality (12) holds unchanged in $\mathbb{R}^d$, the argument of Stam can be run directly in dimension d with no induction at all. This is the presentation adopted by Dembo, Cover and Thomas [16], who obtain the vector case by extending the required tools rather than by reducing to $d = 1$.

The case of equality also differs for the two formulations in dimension d. Equality in the Fisher information inequality forces the equality in the concave form of the EPI to be when X and Y are Gaussian with identical covariance matrices. The equality case for the other formulation is when the two covariance matrices are proportional.

3.2 Gavalakis and Kontoyiannis: Conditions for Equality and Stability

All the proofs discussed so far require for the characterisation of the equality case, in one form or another, an integrability assumption that is not needed to the statement of the inequality itself: a finite second moment for the Fisher information route, or $f \in L^s$ for some $s > 1$ for Lieb's argument. The assumption is used at exactly two places. It is needed first to justify the exchange of differentiation and expectation in de Bruijn's identity, as discussed above, and second to guarantee the continuity of $t \mapsto h(X+\sqrt{t}Z)$ at $t = 0$, without which the integration of the Fisher information inequality from $t = 0$ to $t = \infty$ cannot be closed. Bobkov and Chistyakov [4] exhibited a random variable X with finite differential entropy such that $h(X+Y) = +\infty$ for every independent Y with finite entropy, showing that the finiteness of $h(X)$ alone does not suffice to run the argument.

Gavalakis and Kontoyiannis [1] showed that this pathological behavior is the only obstruction. They proved that assuming the existence of an independent Y with finite entropy such that

$h(X+Y) < \infty$ is sufficient to rule it out, and this minimal condition replaces the finite second moment assumption in all that follows.

Under this assumption alone they prove that $t \mapsto h(X + \sqrt{t}Z)$ is continuous at $t = 0$, and that de Bruijn's identity holds in the form recalled above. Barron's justification of the exchange of derivative and expectation is bypassed by a submodularity inequality for the entropy of sums, which for independent X, Y, Z is

$$h(X+Y+Z) + h(X) \;\leq\; h(X+Y) + h(X+Z),$$

For the inequality itself, no assumption is in fact needed beyond the existence of the entropies involved, since the statement is true whenever $h(X+Y) = +\infty$. The difficulty lies in the equality case, which the earlier proofs could only settle under the assumptions that the pathological examples violate. The main result of [1] is that, for real-valued X and Y with finite differential entropies and no further assumption, equality

$$h\left(\sqrt{\lambda}X + \sqrt{1-\lambda}\,Y\right) = \lambda h(X) + (1-\lambda)h(Y)$$

holds for some $\lambda \in (0,1)$ if and only if X and Y are Gaussian with the same variance. This is the first characterisation of the equality case requiring nothing but the finiteness of the entropies.

These results are established on the real line. The authors observe that their stability argument extends to random vectors through the Lévy-Prokhorov metric, but that doing so presupposes the equality case of the EPI in $\mathbb{R}^d$ under the same minimal assumptions, which their one-dimensional analysis does not provide. Obtaining that d-dimensional characterisation is the object of the section 4.

3.3 Equality, Stability, and Discrete Analogues of the EPI

On $\mathbb{Z}$ neither of the two formulations survives. The entropy power form fails instantly: taking X and Y deterministic gives $H(X) = H(Y) = H(X+Y) = 0$, so that $e^{2H(X+Y)} = 1 < 2 = e^{2H(X)} + e^{2H(Y)}$. The concave form, on the other hand, becomes true whenever it makes sense, since conditioning on one of the variables yields $H(X_1 + X_2) \geq H(X_1 + X_2 \mid X_2) = H(X_1)$ and a convex combination of these inequalities gives the result. The obstruction comes from the fact that there is no real-valued scaling possible for integers in $\mathbb{Z}$.

One answer, used in this report, keeps the concave form and focuses on how much $H(X_1+X_2)$ exceeds $H(X_1)$. Tao showed that for i.i.d. X_1, X_2 on a torsion-free group [18],

$$H(X_1+X_2) \geq H(X_1) + \frac{1}{2}\log 2 - o(1) \quad \text{as } H(X_1) \to \infty,$$

a bound attained by Bernoulli(1/2) variables. The $o(1)$ term cannot be removed in general, so quantifying it became the natural question.

The key tool is to smooth $\mathbb{Z}$-valued variables to make them continuous: if U is uniform on $(0,1)$ and independent of X, $h(X+U) = H(X)$. Comparing $X_1 + X_2$ smoothed by two uniforms to the same sum smoothed by one uniform is the main difficulty. Under log-concavity, Gavalakis bounded this gap of order $O(\log \sigma/\sigma)$ [2], later extended to $\mathbb{Z}^d$ by Fradelizi, Gavalakis and Rapaport [17].

Gavalakis and Kontoyiannis [1] obtained a stability result. They also proved using the continuous result, Cheeger's inequality, and a bound on the Poincare constant of $X + U$, a quantitative version of Tao's inequality for log-concave variables, with equality case attained with discretised Gaussian.

4 Contributions I: The Continuous Setting in $\mathbb{R}^d$

4.1 The Pathological Class $\mathcal{C}_{BC}$ in Dimension d

We show in this section that the obstruction shown by Bobkov and Chistyakov [4], originaly constructed in dimension one, generalizes to $\mathbb{R}^d$.

Definition 4.1. We define the $\mathbb{R}^d$ pathological class as

$$\mathcal{C}_{\text{BC}} := \{X \text{ r.v. in } \mathbb{R}^d : h(X) \text{ finite and } h(X+Y) = \infty \text{ for any independent } Y \text{ with } h(Y) > -\infty\}.$$

Proposition 4.2. *There exists a random variable $X \in \mathbb{R}^d$ such that $-\infty < h(X) < \infty$ and $h(X+Y) = +\infty$ for every independent random variable Y with $h(Y) > -\infty$.*

Proof. We use the same argument used in the proof of [4] for the case $d = 1$.

The result was proved by constructing a random variable X such that $-\infty < h(X) < \infty$ and $h(X + X') = \infty$ for any independent random variable X' with the same distribution. We have to find a probability density with finite entropy whose convolution with itself does not have finite entropy.

The result then follows from the submodularity-for-sum inequality. We suppose by contradiction that $h(X+Y) < \infty$. Then the inequality holds as $h(Y)$ and $h(X+Y)$ are both finite and we have the contradiction:

$$+\infty = h(X+X') \leq h(X+Y+X') \leq 2h(X+Y) - h(Y) < +\infty \tag{13}$$

Considering the random vector $\mathbf{X} = (X_1, \ldots, X_d) \in \mathbb{R}^d$ where all X_i and i.i.d. with the same distribution as the one used in the counterexample in $d = 1$, we notice that $h(\mathbf{X}) = d{\cdot}h(X)$ which is finite and for $\mathbf{X}'$ independent with the same distribution as $\mathbf{X}$, $h(\mathbf{X}+\mathbf{X}') = d{\cdot}h(X+X') = +\infty$.

Therefore, for any independent random variable $\mathbf{Y}$, we deduce $h(\mathbf{X}+\mathbf{Y}) = +\infty$ using the same process.

□

Having shown that the class is non empty, we now show that it is the only obstruction. For any random vector, it either is in the class or behaves nicely under convolution.

Theorem 4.3. *Let X be a random vector in $\mathbb{R}^d$ whose differential entropy exists and is finite. There are only two possibilities:*

1. *$X \in \mathcal{C}_{BC}$*
2. *$h(X+Y) < \infty$ for all independent random vectors $Y \notin \mathcal{C}_{BC}$ with $h(Y) < \infty$.*

See section A.1 for the proof of theorem 4.3.

4.2 Entropy Continuity under Gaussian Perturbation

As a consequence of this dichotomy, we prove the continuity at $t = 0$ of the entropy under gaussian perturbation. This is done under the minimal assumption of x not being in the pathology class exhibited above.

Theorem 4.4 (Entropy continuity under Gaussian perturbation)**.** *Let X be a random vector in $\mathbb{R}^d$ with finite differential entropy, for which there exists an independent random variable Y with finite differential entropy such that $h(X+Y) < \infty$. Then, if $Z \sim \mathcal{N}(0, I_d)$ is independent of X,*

$$\lim_{t \downarrow 0} h(X + \sqrt{t}Z) = h(X).$$

See proof in section A.2.

This result is needed for the justification of De Bruijn's identity and will be used in the next section. The proof relies on the same argument and method as the one dimensional case proved in [1], after verifying that each step and argument extends to $\mathbb{R}^d$.

4.3 De Bruijn's Identity without Moment Assumptions in $\mathbb{R}^d$

Before proving De Bruijn's identity below, we first show that the density of the convolution satisfies the heat equation. It is the starting point, alongside two other technical lemmas, to proving the result.

Lemma 4.5. *Let X be a random vector in $\mathbb{R}^d$. Then:*

$$\frac{\partial}{\partial t} g_t(y) = \frac{1}{2} \Delta_y \, g_t(y), \quad \text{for every } t > 0,\, y \in \mathbb{R}^d \tag{14}$$

Where $\Delta_y \, g_t(y) = \sum_{i=1}^{d} \frac{\partial^2}{\partial y_i^2} g_t(y)$

Theorem 4.6 (De Bruijn's identity)**.** *Let X be a random vector in $\mathbb{R}^d$ with finite differential entropy, for which there exists an independent random vector Y with finite differential entropy such that $h(X+Y) < \infty$. Then, if $Z \sim \mathcal{N}(0, I_d)$ is independent of X,*

$$\frac{d}{dt} h(X + \sqrt{t}Z) = \frac{1}{2} I(X + \sqrt{t}Z), \quad t > 0.$$

The proof of the identity is given in A.3. We finally have all the tools to prove the EPI in $\mathbb{R}^d$ with minimal assumptions.

Theorem 4.7 (EPI in $\mathbb{R}^d$)**.** *Let X, Y be independent random vectors in $\mathbb{R}^d$. Then, for any $\lambda \in (0,1)$,*

$$h(\sqrt{\lambda}X + \sqrt{1-\lambda}\,Y) \geq \lambda h(X) + (1-\lambda) h(Y).$$

If $h(X)$ and $h(Y)$ exist and are finite, there is equality if and only if X and Y are Gaussian with the same covariance matrix.

Remark 4.8. We do not reproduce the proof here, as it follows Stam's classical argument via de Bruijn's identity (Theorem 4.6), adapted to $\mathbb{R}^d$ exactly as in the proof given in [16]. The removal of the hypothesis in the equality case for R^d comes from the removal of these hypothesis in De Bruijn's identity.

4.4 Stability under the Lévy-Prokhorov Metric

We can now give a qualitative stability result to the EPI in dimension d. The proof of the result is a generalisation of the proof for the $\mathbb{R}$ form in [1] and relies on the same technique.

We define the EPI deficit $\delta_{\mathrm{EPI},\lambda}(X,Y)$ by:

$$\delta_{\mathrm{EPI},\lambda}(X,Y) = h(\sqrt{\lambda}X + \sqrt{1-\lambda}\,Y) - \lambda h(X) - (1-\lambda)h(Y)$$

Theorem 4.9. *Let X, Y be independent random vectors in $\mathbb{R}^d$ with finite differential entropies, $-\infty < h(X), h(Y) < \infty$, and $\mathbb{E}[\|X\|^k], \mathbb{E}[\|Y\|^k] < \infty$ for some $k > 1$. Fix $\lambda \in (0,1)$. Then for each $\epsilon > 0$ there is a $\delta > 0$ such that $\delta_{EPI,\lambda}(X,Y) < \delta$ implies*

$$d_{LP}(X, G_1), d_{LP}(Y, G_2) < \epsilon$$

for some Gaussian random vectors G_1, G_2 with the same covariance matrix.

Equivalently, suppose $\lambda \in (0,1)$ and $\{(X_n, Y_n)\}$ is a sequence of pairs of random vectors in $\mathbb{R}^d$ such that, for each n, X_n, Y_n are independent with finite differential entropies. If $\{X_n\}$ and $\{Y_n\}$ have uniformly bounded k-th absolute moments for some $k > 1$ and $\delta_{EPI,\lambda}(X_n, Y_n) \to 0$ as $n \to \infty$, then the limits of any subsequence along which $\{X_n\}$ and $\{Y_n\}$ both converge weakly, are Gaussian with the same covariance matrix.

The proof of this result is given in section A.4.

5 Contributions II: The Discrete Setting in $\mathbb{Z}$

5.1 Approximate EPI for Sums of Log-Concave Variables

We now prove results for the discrete counterpart of the entropy power inequality on $\mathbb{Z}$. As explained above, the natural way to compare $H(S)$ to a continuous entropy is to smooth S with an independent uniform variable U, since $h(S+U) = H(S)$ exactly. Our first result makes this comparison quantitative when S is log-concave: we show that smoothing with m independent uniforms changes the entropy by an error term that vanishes as the variance σ_S^2 grows, for any fixed m. This will be the key technical tool used to prove the discrete entropy power inequality in Theorem 5.2.

Theorem 5.1. *Let $m \geq 1$ and let S be a log-concave discrete random variable on $\mathbb{Z}$ with variance σ_S^2, and let $U^{(m)} := \sum_{i=1}^m U_i$ with U_i i.i.d. uniforms on $(0,1)$ independent of S. Then*

$$h(S + U^{(m)}) = H(S) + o(1),$$

where the $o(1)$-term vanishes as $\sigma_S \to \infty$ for fixed m. More precisely, provided that

$$\sigma_S \;\geq\; \max\Big\{\, 2^{m+2},\; (12e^3)^{5/3} \,\Big\}, \tag{15}$$

we have the explicit bound

$$\left|h(S+U^{(m)}) - H(S)\right| \;\le\; 2^{m+6}\, e^{-\sigma_S^{1/5}} \sigma_S^3 \;+\; \frac{2^{m+2}}{\sigma_S} \log(2^{m+2}\sigma_S) \;+\; \frac{\log \sigma_S^2}{8\sigma_S^2} \;=:\; \varepsilon_m(\sigma_S). \quad (16)$$

Using this approximation, we can now use the continuous EPI to the discrete case: applying it to the smoothed variables $X_i + U_i$ and comparing back to $H(S)$ via Theorem 5.1 yields an approximate EPI for sums of independent, not necessarily identically distributed, log-concave random variables on $\mathbb{Z}$. This was only done before for iid random variables [2].

Theorem 5.2. *Let $X_1, \dots, X_n$ be n independent, not necessarily identically distributed, log-concave random variables on $\mathbb{Z}$ with finite variances $\sigma_1^2, \dots, \sigma_n^2$, and set $\sigma_S^2 := \sum_{i=1}^n \sigma_i^2$. Then*

$$H\Big(\sum_i X_i\Big) \ge \frac{1}{n} \sum_i H(X_i) + \frac{1}{2} \log n - o(1),$$

where the $o(1)$-term vanishes as $\sigma_S \to \infty$ for fixed n. More precisely, provided that $\sigma_S \ge \max\{2^{n+2},\, (12e^3)^{5/3}\}$,

$$H\Big(\sum_i X_i\Big) \ge \frac{1}{n} \sum_i H(X_i) + \frac{1}{2} \log n - \varepsilon_n(\sigma_S),$$

with ε_n as in (16).

5.2 Discrete Stability Result

We conclude this section with a stability result to the discrete entropy power inequality of Theorem 5.2. The strategy follows the same idea as before by smoothing the variables. This allows us to invoke the continuous stability result alongside a result from Yao and Yuan [10].

Theorem 5.3 (Discrete stability)**.** *Let X_1, X_2 be two independent log-concave random variables on $\mathbb{Z}$, of mean μ_i and variance σ_i^2. We define $Z_i^{(\mathbb{Z})}$ as the Gaussian $\mathcal{N}(\mu_i, \sigma_i^2)$ discretized on $\mathbb{Z}$ and the discrete deficit as:*

$$\delta_H(X_1, X_2) := H(X_1 + X_2) - \frac{1}{2} H(X_1) - \frac{1}{2} H(X_2) - \frac{1}{2} \log 2.$$

Then there exist constants $C_1(\sigma_1, \sigma_2), C_2 > 0$ such that:

$$\frac{D(X_1 \| Z_1^{(\mathbb{Z})}) + D(X_2 \| Z_2^{(\mathbb{Z})})}{2} \;+\; \frac{1}{2} \log \frac{\sigma_1^2 + \sigma_2^2}{2\,\sigma_1 \sigma_2} \;\le\; C_1(\sigma_1, \sigma_2)\, \delta_H(X_1, X_2) \;+\; C_2\, \frac{\log \sigma_{\min}}{\sigma_{\min}}.$$

Remark 5.4. The constant $C_1(\sigma_1, \sigma_2)$ is bounded when σ_1 and σ_2 are close to one another. It grows to infinity whenever $\frac{\max(\sigma_i)}{\min(\sigma_i)}$ diverges to infinity but this is not a problem as $\delta_H(X_1, X_2)$ does not converge to 0 anyway in this setting.

6 Discussion and Perspectives

A natural direction for future work is to extend the discrete results of Section 5 from $\mathbb{Z}$ to $\mathbb{Z}^d$. The approximate entropy power inequality of Theorem 5.2 and the stability result of Theorem 5.3 both rely on the technique of smoothing by independent uniform variables. Passing from $\mathbb{Z}$ to $\mathbb{Z}^d$ creates a lot of obstruction: Hoggar's theorem, which guarantees that the sum of independent log-concave random variables on $\mathbb{Z}$ remains log-concave, is specific to dimension one and has no known analogue for $d \geq 2$ under the natural generalization of log-concavity. Fradelizi, Gavalakis and Rapaport [17] avoid this issue by working with a broader class of probability mass functions with better control which is preserved (up to constants) under convolution. Extending Theorems 5.2 and 5.3 to $\mathbb{Z}^d$ would likely require adopting a similar relaxed notion of log-concavity rather than working with the direct multivariate generalization of 11.

7 Conclusion

7.1 scientific conclusion

This report extended the recent results of Gavalakis and Kontoyiannis on the equality case and stability of Shannon's entropy power inequality to random vectors in $\mathbb{R}^d$ through the generalisation of De Bruijn's identity, and obtained a discrete counterpart of these results on $\mathbb{Z}$.

On the continuous side, we showed that the pathological behaviour exhibited by the Bobkov-Chistyakov counterexample is, as in dimension one, the only obstruction to the behavior of the differential entropy under convolution and Gaussian perturbation in $\mathbb{R}^d$. This allowed us to establish the continuity of $t \mapsto h(X + \sqrt{t}Z)$ at $t = 0$ and de Bruijn's identity under no assumption beyond the finiteness of the relevant entropies, therefore removing the second-moment assumptions present in all previous vector-valued proofs. this yields a characterization of the equality case in the vector EPI, and a qualitative stability result under the Lévy-Prokhorov metric.

On the discrete side, we obtained a quantitative approximation of the discrete entropy of a log-concave sum, which we used to derive an approximate entropy power inequality for sums of independent, not necessarily identically distributed, log-concave random variables on $\mathbb{Z}$, together with an associated stability result. These results extend the work of Tao and of Gavalakis and Kontoyiannis beyond the identically distributed setting.

7.2 general conclusion

This internship was a superb experience. It allowed me to immerse myself in a completely different culture for several months. The department was very dedicated to making my stay as good as possible: I was given an office to share with two PhD students, as well as access to all university facilities. I learned a lot mathematically through all the discussions I had in the department, the weekly seminars, and the opportunity to attend the 4th Information Theory Colloquium in Cambridge. I would recommend the city and its university to anyone willing to learn, or simply looking to visit a beautiful place and enjoy a wonderful time.

A Technical Lemmas and proofs of the theorems

Note that the proofs of the results in the first two subsections of this appendix follow exactly the same method as in the $d = 1$ case. The main challenge here was to carefully verify that each argument transposes well to the high-dimensional setting without any constraints.

A.1 Proof of theorem 4.3

Lemma A.1. *Let X be a random vector in $\mathbb{R}^d$ with entropy $-\infty < h(X) < +\infty$ such that there exists a random variable Y independent of X, with finite entropy and with $h(X+Y) < +\infty$. Then $h(X+Z) < +\infty$ for any Z independent with finite second moment.*

Proof. Let's first consider X,X' i.i.d., then we have:

$$h(X+X') \leq h(X+Y+X') \leq 2h(X+Y) - h(Y) < +\infty \tag{17}$$

We assume that $\|X\|$ is not bounded, because otherwise $\mathbb{E}[\|X\|^2] < \infty$, $\mathbb{E}[\|X+Z\|^2] < +\infty$ and therefore

$$h(X+Z) \leq \frac{1}{2}\log((2\pi e)^d |\mathrm{cov}(X+Z)|) < +\infty$$

Because the determinant of the covariance matrix is finite.

Let $I_n := \mathbb{I}_{\{\|X\| \leq n\}}$. As $\|X\|$ is not bounded, there exists an n such that $p_n = \mathbb{P}(\|X\| \leq n) \in (0,1)$.

We also let X_n be distributed as X conditioned on $\{I_n = 1\}$. Writing the mutual information between I_n and X, we observe:

$$\begin{aligned} I(X; I_n) &= H(I_n) - H(I_n|X) \\ &= h(X) - h(X|I_n) \end{aligned} \tag{18}$$

And as I_n is a function of X, $h(I_n|X) = 0$ and we deduce $h(X|I_n) = h(X) - H(I_n)$ and $h(X|I_n) \in \mathbb{R}$.

Therefore, as $h(X|I_n) = p_n h(X_n) + (1-p_n)h(X|I_n = 0)$ and $h(X|I_n = 0) \leq h(X) < +\infty$, we see that $h(X_n) > -\infty$.

Since conditioning reduces the entropy, we also have:

$$\begin{aligned} +\infty > h(X+X') \geq h(X+X'|I_n) &= p_n h(X_n + X') + (1-p_n)h(X+X'|I_n = 0) \\ &\geq p_n h(X_n + X') + (1-p_n)h(X'|I_n = 0) \\ &= p_n h(X_n + X') + (1-p_n)h(X') \end{aligned}$$

And therefore $h(X_n + X') < +\infty$. Finaly, using again the submodularity-for-sum inequality, we have:

$$h(X'+Z) \leq h(X'+X_n+Z) \leq h(X'+X_n) + h(X_n+Z) - h(X_n) < +\infty \tag{19}$$

Where $h(X_n+Z) < +\infty$ because $\|X_n\|$ is bounded and therefore $\mathbb{E}[\|X_n\|^2]$ is finite, as well as $\mathbb{E}[\|X_n+Z\|^2]$ which implies that $h(X_n+Z)$ is finite.

□

Theorem 4.3. *Let X be a random vector with $-\infty < h(X) < +\infty$, then either:*

1. $X \in C_{BC}$
2. $h(X+Y) < +\infty$ *for all* $Y \notin C_{BC}$ *with* $h(Y) < +\infty$

Proof. Let X be a random variable with $-\infty < h(X) < +\infty$. We only need to prove that if there exists a random vector W with finite entropy such that $h(X+W) < +\infty$ and if Y, V are two independent random vectors, independent from X, W, with finite differential entropies, such that $h(Y+V) < \infty$, then:

$$h(X+Y) < \infty$$

Let Z be a standard Gaussian (any random vector with finite second moment works). Then:

$$h(X+Y) \leq h(X+Z+Y) \leq h(X+Z) + h(Z+Y) - h(Z) \tag{20}$$

Where the finiteness of the last two entropies follows by Lemma A.1.

□

A.2 Proof of theorem 4.4

Before proceeding with the next lemma, it is worth noting that extending it to $\mathbb{R}^d$ only requires substituting the absolute value with the Euclidean norm. We have systematically verified that every step of the argument remains valid and rigorous under this multidimensional generalization without requiring additional assumptions.

Lemma A.2. *Let X be a random vector in $\mathbb{R}^d$ with finite entropy, Z a standard gaussian vector independent of X, and $t_0 > 0$. Assume that $h(X + \sqrt{t_0}Z) < \infty$. Then:*

$$\lim_{n \to \infty} h(X + \sqrt{t_0}Z \mid \|X\| \leq n) = h(X + \sqrt{t_0}Z)$$

Proof. Write f for the density of X and $g_{t_0}^{(n)}$ for the density of $X + \sqrt{t_0}Z$ conditioned on $\{\|X\| \leq n\}$. Then

$$g_{t_0}^{(n)} = \frac{1}{p_n} \int_{\|X\| \leq n} f(x) \varphi_{t_0}(z - x)$$

Where φ_t denotes the $\mathcal{N}(0, I_d)$ density, and $p_n = \mathbb{P}(\|X\| \leq n) \to 1$ as $n \to \infty$. We also have $p_n g_{t_0}^{(n)} \uparrow g_{t_0}$ as $n \to \infty$.

We can write

$$
\begin{aligned}
&h(X+\sqrt{t_0}Z|\,\|X\|\le n)\\
&=-\int_{0\le g_{t_0}\le\frac{1}{e}} g_{t_0}^{(n)}(z)\log g_{t_0}^{(n)}(z)dz-\int_{\frac{1}{e}<g_{t_0}\le 1} g_{t_0}^{(n)}(z)\log g_{t_0}^{(n)}(z)dz-\int_{1<g_{t_0}} g_{t_0}^{(n)}(z)\log g_{t_0}^{(n)}(z)dz\\
&=-\frac{1}{p_n}\int_{0\le g_{t_0}\le\frac{1}{e}} p_n g_{t_0}^{(n)}(z)\log p_n g_{t_0}^{(n)}(z)dz-\frac{1}{p_n}\int_{\frac{1}{e}<g_{t_0}\le 1} p_n g_{t_0}^{(n)}(z)\log p_n g_{t_0}^{(n)}(z)dz\\
&\quad-\frac{1}{p_n}\int_{1<g_{t_0}} p_n g_{t_0}^{(n)}(z)\log p_n g_{t_0}^{(n)}(z)dz+o(1)
\end{aligned}
$$

In the range of the first integral, we notice that $-x\log x$ is increasing for $x\le\frac{1}{e}$ and therefore by monotone convergence, as $-p_n g_{t_0}^{(n)}\log(p_n g_{t_0}^{(n)})\uparrow -g_{t_0}\log g_{t_0}$, we have the convergence of the integrals.

For the second integral, denoting $A:=\{\frac{1}{e}<g_{t_0}\le 1\}$, we notice that $1=\int_{\mathbb{R}^d} g_{t_0}\ge\int_A g_{t_0}\ge\int_A\frac{1}{e}=\frac{1}{e}\cdot\mathrm{Leb}(A)$. In addition, $-x\log x$ is bounded for $0\le x\le 1$ so by bounded convergence, we have

$$
-\int_{\frac{1}{e}<g_{t_0}\le 1} p_n g_{t_0}^{(n)}(z)\log(p_n g_{t_0}^{(n)}(z))dz\to-\int_{\frac{1}{e}<g_{t_0}\le 1} g_{t_0}(z)\log(g_{t_0}(z))dz
$$

Finally, writing the third term as

$$
\begin{aligned}
&-\frac{1}{p_n}\int_{1<g_{t_0}} p_n g_{t_0}^{(n)}(z)\log(p_n g_{t_0}^{(n)}(z))dz\\
&=-\frac{1}{p_n}\int_{1<g_{t_0}} \mathbb{I}_{\{p_n g_{t_0}^{(n)}(z)\le 1\}} p_n g_{t_0}^{(n)}(z)\log(p_n g_{t_0}^{(n)}(z))dz-\frac{1}{p_n}\int_{1<g_{t_0}} \mathbb{I}_{\{p_n g_{t_0}^{(n)}(z)>1\}} p_n g_{t_0}^{(n)}(z)\log(p_n g_{t_0}^{(n)}(z))dz
\end{aligned}
$$

We notice that the integrand in the first term converges to 0 pointwise and using again the bounded convergence theorem, as the set is of Lebesgue measure at most 1, we have

$$
-\frac{1}{p_n}\int_{1<g_{t_0}} \mathbb{I}_{\{p_n g_{t_0}^{(n)}(z)\le 1\}} p_n g_{t_0}^{(n)}(z)\log(p_n g_{t_0}^{(n)}(z))dz\to 0
$$

On the other hand, for the second integral, as $x\log x$ is non-decreasing for for $x>1>\frac{1}{e}$ and monotone convergence again,

$$
\frac{1}{p_n}\int_{1<g_{t_0}} \mathbb{I}_{\{p_n g_{t_0}^{(n)}(z)>1\}} p_n g_{t_0}^{(n)}(z)\log(p_n g_{t_0}^{(n)}(z))dz\uparrow\int_{1<g_{t_0}} g_{t_0}(z)\log(g_{t_0}(z))dz
$$

This concludes the proof.

□

Theorem 4.4 (Entropy continuity under Gaussian perturbation)**.** *Let X be a random vector in $\mathbb{R}^d$ with finite differential entropy, for which there exists an independent random variable Y with finite differential entropy such that $h(X+Y)<\infty$. Then, if $Z\sim\mathcal{N}(0,I_d)$ is independent of X,*

$$
\lim_{t\downarrow 0} h(X+\sqrt{t}Z)=h(X)
$$

Proof. The proof follows exactly the same strategy as the one-dimensional case in [1], with every step transposing to $\mathbb{R}^d$ by replacing $|\cdot|$ with $\|\cdot\|$, including the use of Barron's Lemma 2.1 [5]. We therefore do not reproduce the full proof here.

□

A.3 Proof of theorem 4.6

Unlike the preceding results, the proofs of the following lemmas diverge from their one-dimensional counterparts. Extending these properties to the multidimensional setting required different techniques, most importantly in the proof of the final lemma.

Lemma 4.5. *Let X be a random vector in $\mathbb{R}^d$. Then:*

$$\frac{\partial}{\partial t} g_t(y) = \frac{1}{2} \Delta_y \, g_t(y), \quad \textit{for every } t > 0,\ y \in \mathbb{R}^d \tag{14}$$

Where $\Delta_y \, g_t(y) = \sum_{i=1}^{d} \frac{\partial^2}{\partial y_i^2} g_t(y)$

Proof. **Step 1** $\frac{\partial}{\partial t}\varphi_t(y) = \Delta_y \, \varphi_t(y)$

The first step is to verify that the integrand verifies the heat equation itself. Let $g_t = f * \varphi_t$. Then

$$g_t(y) = \mathbb{E}[\varphi_t(y - X)] = \int_{\mathbb{R}^d} \varphi_t(y - x) f(x) \, dx$$

We notice that:

$$\begin{aligned}\frac{\partial}{\partial t}\varphi_t(y) = \frac{\partial}{\partial t}\left(\frac{1}{(2\pi t)^{d/2}} \exp(-\frac{\|y\|^2}{2t})\right) &= \left(\frac{(2\pi t)^{-d/2}\|y\|^2}{2t^2} - \frac{\pi d}{(2\pi t)^{\frac{d+2}{2}}}\right) \exp(-\frac{\|y\|^2}{2t}) \\ &= \left(\frac{\|y\|^2}{2t^2} - \frac{d}{2t}\right) \varphi_t(y)\end{aligned}$$

And

$$\frac{\partial^2}{\partial y_i^2}\varphi_t(y) = \left(\frac{|y_i|^2}{t^2} - \frac{1}{t}\right) \varphi_t(y)$$

Which makes

$$\Delta_y \, \varphi_t(y) = \sum_{i=1}^{d} \frac{\partial^2}{\partial y_i^2}\varphi_t(y) = \left(\frac{\|y\|^2}{t^2} - \frac{d}{t}\right) \varphi_t(y) = 2 \cdot \frac{\partial}{\partial t}\varphi_t(y) \tag{21}$$

Step 2 $\frac{\partial}{\partial t}\mathbb{E}[\varphi_t(y - X)] = \mathbb{E}[\frac{\partial}{\partial t}\varphi_t(y - X)]$

Let's fix $0 < a < t < b$.

$$\frac{\partial}{\partial t}\varphi_t(z) = \frac{1}{(2\pi t)^{d/2}}\frac{1}{2t}\left(\frac{\|z\|^2}{t} - d\right)\exp\left(-\frac{\|z\|^2}{2t}\right) \quad \text{where } z = y - x$$

Using the fact that $\sup_{u>0} u \cdot \exp(-u/2) \leq K < +\infty$, we have

$$\left|\frac{\partial}{\partial t}\varphi_t(z)\right| \leq \frac{1}{(2\pi t)^{d/2}}\frac{1}{2t}K' \leq \frac{1}{(2\pi a)^{d/2}}\frac{1}{2a}K' = K'' \quad \text{for all } z \in \mathbb{R}^d,\, t \in (a,b)$$

And we apply the dominated convergence theorem to exchange the differentiation and expectation as K'' is integrable against $f(x)\lambda(x)$.

Step 3: $\frac{\partial^2}{\partial y_i^2}\mathbb{E}[\varphi_t(y-X)] = \mathbb{E}[\frac{\partial^2}{\partial y_i^2}\varphi_t(y-X)]$

First we have that

$$\left|\frac{\partial}{\partial y_i}\varphi_t(y-x)\right| = \frac{|z_i|}{t}\varphi_t(z) = \quad \text{where } z = y - x$$

Which is integrable, so with the dominated convergence theorem we have $\frac{\partial}{\partial y_i}\mathbb{E}[\varphi_t(y-X)] = \mathbb{E}[\frac{\partial}{\partial y_i}\varphi_t(y-X)]$.

And as $\frac{\partial^2}{\partial y_i^2}\varphi_t(z) = \left(\frac{|z_i|^2}{t^2} - \frac{1}{t}\right)\varphi_t(z)$ is dominated for the same reasons as above with $\frac{\partial}{\partial t}\varphi_t(z)$, we have:

$$\Delta_y\,\mathbb{E}\left[\varphi_t(y-X)\right] = \sum_{i=1}^{d}\frac{\partial^2}{\partial y_i^2}\mathbb{E}[\varphi_t(y-X)] = \mathbb{E}\left[\sum_{i=1}^{d}\frac{\partial^2}{\partial y_i^2}\varphi_t(y-X)\right] = \mathbb{E}\left[\Delta_y\,\varphi_t(y-X)\right]$$

Which concludes the proof.

□

Lemma A.3. *Let X be a random vector with density f, finite differential entropy and for which there exists an independent random variable Y with finite differential entropy itself such that $h(X+Y) < \infty$. Then, writing $g_t = f * \varphi_t$, we have:*

$$\frac{\partial}{\partial t}h(X+\sqrt{t}Z) = -\int\left(\frac{\partial}{\partial t}g_t(y)\right)\log g_t(y)dy \tag{22}$$

Proof. We have $h(X+\sqrt{t_0}Z) < +\infty$ for every $t_0 > 0$. We have to show that $\frac{\partial}{\partial t}(g_t \log g_t)$ is dominated by an integrable function uniformly in a neighborhood of t. We then fix $0 < a < t < t_0/2$.

First we notice that:

$$\begin{aligned}\frac{\partial}{\partial t}\varphi_t(y) = \frac{\partial}{\partial t}\left(\frac{1}{(2\pi t)^{d/2}}\exp(-\frac{\|y\|^2}{2t})\right) &= \left(\frac{(2\pi t)^{-d/2}\|y\|^2}{2t^2} - \frac{\pi d}{(2\pi t)^{\frac{d+2}{2}}}\right)\exp(-\frac{\|y\|^2}{2t}) \\ &= 2^{d/2}\left(\frac{\|y\|^2}{2t^2} - \frac{d}{2t}\right)\exp\left(-\frac{\|y\|^2}{4t}\right)\varphi_{2t}(y)\end{aligned}$$

Moreover,

$$\frac{\varphi_{2t}(y)}{\varphi_{t_0}(y)} = \left(\frac{t_0}{2t}\right)^{d/2} \exp\left(-\frac{\|y\|^2}{2}(\frac{1}{2t} - \frac{1}{t_0})\right) \leq \left(\frac{t_0}{2t}\right)^{d/2} \leq \left(\frac{t_0}{2a}\right)^{d/2}$$

And as $\sup_{u\geq 0} u\exp(-u/4t)$ et $\sup_{u\geq 0}\exp(-u/4t)$ are controlled, we have:

$$\begin{aligned}\left|\frac{\partial}{\partial t} g_t(y)\right| = \left|\frac{\partial}{\partial t}\mathbb{E}[\varphi_t(y-X)]\right| &\leq \mathbb{E}\left[2^{d/2}\left|\frac{\|y-X\|^2}{2t^2} - \frac{d}{2t}\right|\exp\left(-\frac{\|y-X\|^2}{4t}\right)\varphi_{2t}(y-X)\right] \\ &\leq \mathbb{E}\left[C_{a,t_0}\varphi_{t_0}(y-X)\right] = C_{a,t_0} g_{t_0}(y)\end{aligned}$$

We have to dominate $|\log g_t(y)|$ now. First we notice that

$$g_t(y) \leq \frac{1}{(2\pi t)^{d/2}} \leq \frac{1}{(2\pi a)^{d/2}}$$

$$g_t(y) = \mathbb{E}\left[\frac{1}{(2\pi t)^{d/2}}\exp\left(-\frac{\|y-X\|^2}{2t}\right)\right] \geq \mathbb{E}\left[K_{t_0,a}\frac{1}{(2\pi a)^{d/2}}\exp\left(-\frac{\|y-X\|^2}{2a}\right)\right] = K_{t_0,a} g_a(y)$$

Therefore,

$$|\log g_t(y)| \leq \left|\log\left(\frac{1}{(2\pi a)^{d/2}}\right)\right| + |\log K_{t_0,a} g_a(y)|$$

$$|\log g_t(y)| \leq \frac{1}{(2\pi a)^{d/2}} + K_{t_0,a} + |\log g_a(y)| \leq \frac{1}{(2\pi a)^{d/2}} + K_{t_0,a} + |\log g_{t_0}(y)| + \left|\log\frac{g_a(y)}{g_{t_0}(y)}\right|$$

Combining with the bound on $\left|\frac{\partial}{\partial t} g_t(y)\right|$ and writing $K'_{t_0,a} = \frac{1}{(2\pi a)^{d/2}} + K_{t_0,a}$, we get

$$\left|\frac{\partial}{\partial t} g_t(y)\log g_t(y)\right| \leq C'_{t_0,a}\, g_{t_0}(y) + g_{t_0}(y)|\log g_{t_0}(y)| + g_{t_0}(y)\left|\log\frac{g_{t_0}(y)}{g_a(y)}\right|$$

The first term in the right-hand side is clearly integrable. The second term is integrable because of $h(X+\sqrt{t_0}Z) < \infty$. To see that the third term is integrable, we observe first that by the data processing inequality for relative entropy,

$$D(g_{t_0}\|g_a) = D(X+\sqrt{t_0}Z\|X+\sqrt{a}Z) \leq D((X,\sqrt{t_0}Z)\|(X,\sqrt{a}Z)) = D(\sqrt{t_0}Z\|\sqrt{a}Z) < \infty$$

where the finiteness of the last relative entropy follows by simple direct calculation. This implies that the integral of $g_{t_0}\log[g_{t_0}/g_a]$ is finite, and Lebesgue integrability implies that the last term is also integrable.

Thus, $\left|\frac{\partial}{\partial t} g_t(y)\log g_t(y)\right|$ is bounded uniformly in $t \in (a, t_0/2)$ by the right-hand side of the above bound, which is integrable.

By the mean value theorem and dominated convergence we may exchange differentiation and expectation to obtain

$$\begin{aligned}\frac{\partial}{\partial t} h(X+\sqrt{t}Z) &= -\int \frac{\partial}{\partial t}(g_t(y)\log g_t(y))\,dy \\ &= -\int \frac{\partial}{\partial t} g_t(y)\,dy - \int\left(\frac{\partial}{\partial t} g_t(y)\right)\log g_t(y)\,dy \\ &= -\frac{\partial}{\partial t}\int g_t(y)\,dy - \int\left(\frac{\partial}{\partial t} g_t(y)\right)\log g_t(y)\,dy \\ &= -\int\left(\frac{\partial}{\partial t} g_t(y)\right)\log g_t(y)\,dy,\end{aligned}$$

where $\int \frac{\partial}{\partial t} g_t(y)\,dy = \frac{\partial}{\partial t}\int g_t(y)\,dy$.

□

Lemma A.4. *Let $X \in \mathbb{R}^d$ be a random variable with density f, finite differential entropy, and for which there exists an independent random variable Y with finite differential entropy itself such that $h(X+Y)$ is finite. Then,*

$$I(X+\sqrt{t}Z) = -\int_{\mathbb{R}^d} (\Delta_y\, g_t(y)) \log g_t(y)\, dy, \quad t>0, \tag{23}$$

Proof. First, because of Lemma 4.5 and Lemma A.3, we see that $-(\Delta g_t(y)) \log g_t(y)$ is integrable. The Fisher information $I(X+\sqrt{t}Z)$ is finite, which means that $\|\nabla g_t(y)\|^2/g_t(y)$ is integrable. We want to prove that:

$$\int_{\mathbb{R}^d} \frac{\|\nabla g_t(y)\|^2}{g_t(y)}\, dy = \int_{\mathbb{R}^d} -(\Delta_y g_t(y)) \log g_t(y)\, dy \tag{24}$$
$$\sum_{i=1}^{d} \int_{\mathbb{R}^d} \frac{\left|\frac{\partial}{\partial y_i} g_t(y)\right|^2}{g_t(y)}\, dy = -\sum_{i=1}^{d} \int_{\mathbb{R}^d} \frac{\partial^2 g_t(y)}{\partial y_i^2} \log g_t(y)\, dy$$

So it would be sufficient to prove that $\forall i = 1,\dots,d$, we have $\int_{\mathbb{R}^d} \frac{\left|\frac{\partial}{\partial y_i} g_t(y)\right|^2}{g_t(y)}\, dy = -\int_{\mathbb{R}^d} \frac{\partial^2 g_t(y)}{\partial y_i^2} \log g_t(y)\, dy$

Let i be fixed. As the integrands in all the integrals are integrable across all coordinates, using fubini we have to show that:

$$\int_{\mathbb{R}^{d-1}} \left(\int_{\mathbb{R}} \frac{\left|\frac{\partial}{\partial y_i} g_t(y)\right|^2}{g_t(y)}\, dy_i \right) d\hat{y} = -\int_{\mathbb{R}^{d-1}} \left(\int_{\mathbb{R}} \frac{\partial^2 g_t(y)}{\partial y_i^2}\, \log g_t(y)\, dy_i \right) d\hat{y}$$

And therefore we only need to show that for almost every $y_1,\dots,y_{i-1},y_{i+1},\dots,y_d \in \mathbb{R}$,

$$\int_{\mathbb{R}} \frac{\left|\frac{\partial}{\partial s} u(s)\right|^2}{u(s)}\, ds = -\int_{\mathbb{R}} \frac{\partial^2 u(s)}{\partial s^2}\, \log u(s)\, ds \tag{25}$$

Where $u(s) := g_t(y_1,\dots,y_{i-1},s,y_{i+1},\dots,y_d)$. We then use the same technique as Barron did in [5] with absolute continuity.

$$\frac{\partial}{\partial s}\left[\left(\frac{\partial}{\partial s} u(s) \right) \log u(s) \right] = \left(\frac{\partial^2}{\partial s^2} u(s) \right) \log u(s) + \frac{\left|\frac{\partial}{\partial s} u(s)\right|^2}{u(s)} \tag{26}$$

Hence $\left(\frac{\partial}{\partial s} u(s)\right) \log u(s)$ is absolutely continuous as the derivative on the right side is integrable and exists everywhere. Integrating on both sides between R and $-R$, we get:

$$\left[\left(\frac{\partial}{\partial s} u(s) \right) \log u(s) \right]_{-R}^{R} = \int_{-R}^{R} \left(\frac{\partial^2}{\partial s^2} u(s) \right) \log u(s)\, ds + \int_{-R}^{R} \frac{\left|\frac{\partial}{\partial s} u(s)\right|^2}{u(s)}\, ds$$

And writing the left side as $\left|\left(\frac{\partial}{\partial s} u(s)\right) \log u(s)\right| = \frac{\left|\frac{\partial u(s)}{\partial s}\right|}{\sqrt{u(s)}} \left|2\sqrt{u(s)} \log \sqrt{u(s)}\right|$, we notice that the square of $\left|\frac{\partial u(s)}{\partial s}\right| / \sqrt{u(s)}$ is integrable, therefore it converges to 0 as $|s| \to \infty$ and $\left|2\sqrt{u(s)} \log \sqrt{u(s)}\right|$ is also integrable and therefore it also converges to 0 as $|s| \to 0$.

We then let $R \to \infty$ and we have

$$\int_{\mathbb{R}} \left(\frac{\partial^2}{\partial s^2} u(s) \right) \log u(s)\, ds = -\int_{\mathbb{R}} \frac{\left| \frac{\partial}{\partial s} u(s) \right|^2}{u(s)}\, ds \tag{27}$$

Which concludes the proof.

□

Theorem 4.6 (De Bruijn's identity)**.** *Let X be a random vector in $\mathbb{R}^d$ with finite differential entropy, for which there exists an independent random vector Y with finite differential entropy such that $h(X+Y) < \infty$. Then, if $Z \sim \mathcal{N}(0, I_d)$ is independent of X,*

$$\frac{d}{dt} h(X + \sqrt{t}Z) = \frac{1}{2} I(X + \sqrt{t}Z), \quad t > 0.$$

Proof. This follows directly by combining Lemma 4.5, Lemma A.3 and Lemma A.4. Indeed, by Lemma A.3,

$$\frac{d}{dt} h(X + \sqrt{t}Z) = -\int \left(\frac{\partial}{\partial t} g_t(y) \right) \log g_t(y)\, dy.$$

By Lemma 4.5, $\frac{\partial}{\partial t} g_t = \frac{1}{2} \Delta g_t$, so the right-hand side equals $-\frac{1}{2} \int (\Delta g_t(y)) \log g_t(y)\, dy$, which by Lemma A.4 equals $\frac{1}{2} I(X + \sqrt{t}Z)$. □

A.4 Proof of theorem 4.9

Theorem 4.9. *Let X, Y be independent random vectors in $\mathbb{R}^d$ with finite differential entropies, $-\infty < h(X), h(Y) < \infty$, and $\mathbb{E}[\|X\|^k], \mathbb{E}[\|Y\|^k] < \infty$ for some $k > 1$. Fix $\lambda \in (0,1)$. Then for each $\epsilon > 0$ there is a $\delta > 0$ such that $\delta_{EPI,\lambda}(X, Y) < \delta$ implies*

$$d_{LP}(X, G_1), d_{LP}(Y, G_2) < \epsilon$$

for some Gaussian random vectors G_1, G_2 with the same covariance matrix.

Equivalently, suppose $\lambda \in (0,1)$ and $\{(X_n, Y_n)\}$ is a sequence of pairs of random vectors in $\mathbb{R}^d$ such that, for each n, X_n, Y_n are independent with finite differential entropies. If $\{X_n\}$ and $\{Y_n\}$ have uniformly bounded k-th absolute moments for some $k > 1$ and $\delta_{EPI,\lambda}(X_n, Y_n) \to 0$ as $n \to \infty$, then the limits of any subsequence along which $\{X_n\}$ and $\{Y_n\}$ both converge weakly, are Gaussian with the same covariance matrix.

Proof. **Part 1: The sequential statement implies the original.**

First, we prove that the sequential statement of the theorem implies the original one. Let us suppose by contradiction that this implication does not hold.

For a constant $c > 0$, we define the class of pairs of random vectors:

$$\mathcal{C}_{k,c} := \left\{ (X, Y) : X, Y \text{ independent in } \mathbb{R}^d,\ -\infty < h(X), h(Y) < \infty,\ \mathbb{E}[\|X\|^k], \mathbb{E}[\|Y\|^k] < c \right\}.$$

By assumption, there exists $\epsilon > 0$ such that for all $\delta > 0$, we can find $(X, Y) \in \mathcal{C}_{k,c}$ with $\delta_{\mathrm{EPI},\lambda}(X, Y) < \delta$, while for all Gaussian random vectors G_1, G_2 with the same covariance matrix, we have:

$$d_{LP}(X, G_1) \geq \epsilon \quad \text{or} \quad d_{LP}(Y, G_2) \geq \epsilon.$$

Setting $\delta = 1/n$ for each $n \geq 1$, we obtain a sequence of pairs $(X_n, Y_n) \in \mathcal{C}_{k,c}$ satisfying $\delta_{\mathrm{EPI},\lambda}(X_n, Y_n) \to 0$ as $n \to \infty$, such that for all n and all matching Gaussian vectors G_1, G_2:

$$d_{LP}(X_n, G_1) \geq \epsilon \quad \text{or} \quad d_{LP}(Y_n, G_2) \geq \epsilon.$$

Since $(X_n, Y_n) \in \mathcal{C}_{k,c}$, the sequences $\{X_n\}$ and $\{Y_n\}$ have uniformly bounded k-th moments for some $k > 1$. Therefore, the sequence of joint distributions is tight. By the Prokhorov theorem, there exists a subsequence (X_{n_i}, Y_{n_i}) that converges weakly to a limit (X_∞, Y_∞):

$$(X_{n_i}, Y_{n_i}) \xrightarrow[i\to\infty]{d} (X_\infty, Y_\infty).$$

Since $\delta_{\mathrm{EPI},\lambda}(X_n, Y_n) \to 0$ and the sequential statement holds, the limit (X_∞, Y_∞) must be a pair of Gaussian vectors with the same covariance matrix. This contradicts the fact that $d_{LP}(X_{n_i}, G_1) \geq \epsilon$ or $d_{LP}(Y_{n_i}, G_2) \geq \epsilon$ for all matching Gaussian pairs. Thus, the sequential statement indeed implies the original one.

Part 2: The sequential statement holds.

We argue by contradiction. Suppose that the sequential statement is false. Then there exist sequences of independent pairs (X_n, Y_n) in $\mathbb{R}^d$ with uniformly bounded k-th moments for some $k > 1$ such that $\delta_{\mathrm{EPI},\lambda}(X_n, Y_n) \to 0$ as $n \to \infty$, but no subsequence converges weakly to Gaussian limits with matching covariance matrices.

By tightness, there exists a subsequence $(X_j, Y_j)_j$ converging weakly to limits:

$$X_j \xrightarrow{d} X_\infty \quad \text{and} \quad Y_j \xrightarrow{d} Y_\infty,$$

where X_∞ and Y_∞ are not Gaussian vectors with the same covariance matrix.

Now, we introduce the perturbed variables:

$$\tilde{X}_j := \frac{1}{\sqrt{2}}(X_j + Z) \quad \text{and} \quad \tilde{Y}_j := \frac{1}{\sqrt{2}}(Y_j + Z'),$$

where $Z, Z' \sim \mathcal{N}(0, I_d)$ are independent of everything else. The density of $\tilde{X}_j$ is given by:

$$\begin{aligned} f_{\tilde{X}_j}(y) &= (\sqrt{2})^d \cdot \mathbb{E}\left[f_Z\left(\sqrt{2}y - X_j\right)\right] \\ &= \mathbb{E}\left[\frac{1}{\pi^{d/2}} \exp\left(-\left\|y - \frac{X_j}{\sqrt{2}}\right\|^2\right)\right]. \end{aligned}$$

Since $f_{\tilde{X}_j}(y)$ is the expectation of a continuous and bounded function of X_j, it converges pointwise to the density $f_{\tilde{X}_\infty}(y)$ of $\tilde{X}_\infty$ as $j \to \infty$ (and similarly for the density of $\tilde{Y}_j$ to that of $\tilde{Y}_\infty$). In addition, the densities of $\tilde{X}_j$, $\tilde{Y}_j$, $\tilde{X}_\infty$, and $\tilde{Y}_\infty$ are all uniformly bounded:

$$f_{\tilde{X}_j}(y),\ f_{\tilde{Y}_j}(y),\ f_{\tilde{X}_\infty}(y),\ f_{\tilde{Y}_\infty}(y) \leq \frac{1}{\pi^{d/2}}.$$

Furthermore, by the uniform k-th moment assumption and Minkowski's inequality, we have:

$$\mathbb{E}\left[\|\tilde{X}_j\|^k\right] < C_k \quad \text{and} \quad \mathbb{E}\left[\|\tilde{Y}_j\|^k\right] < C_k$$

for some constant C_k depending on the moment bounds of X_n, Y_n and k.

We also notice that, for any truncation parameter $M > 0$:

$$\mathbb{E}\left[\|\tilde{X}_\infty\|^k; \|\tilde{X}_\infty\| \le M\right] = \lim_{j\to\infty} \mathbb{E}\left[\|\tilde{X}_j\|^k; \|\tilde{X}_j\| \le M\right] \le C_k.$$

Taking $M \to \infty$, we see that $\tilde{X}_\infty$ and $\tilde{Y}_\infty$ satisfy the k-th moment condition. Using these density bounds, pointwise convergence, and uniform moment bounds, we conclude the proof in the same manner as the $d = 1$ case. □

A.5 Proof of theorem 5.2

Theorem 5.1. *Let $m \ge 1$ and let S be a log-concave discrete random variable on $\mathbb{Z}$ with variance σ_S^2, and let $U^{(m)} := \sum_{i=1}^m U_i$ with U_i i.i.d. uniforms on $(0,1)$ independent of S. Then*

$$h(S + U^{(m)}) = H(S) + o(1),$$

where the $o(1)$-term vanishes as $\sigma_S \to \infty$ for fixed m. More precisely, provided that

$$\sigma_S \;\ge\; \max\Big\{\, 2^{m+2},\; (12e^3)^{5/3} \,\Big\}, \tag{15}$$

we have the explicit bound

$$\Big|h(S + U^{(m)}) - H(S)\Big| \;\le\; 2^{m+6}\, e^{-\sigma_S^{1/5}} \sigma_S^3 \;+\; \frac{2^{m+2}}{\sigma_S} \log(2^{m+2}\sigma_S) \;+\; \frac{\log \sigma_S^2}{8\sigma_S^2} \;=:\; \varepsilon_m(\sigma_S). \tag{16}$$

Proof. Assume without loss of generality that $\mathbb{E}[S] = 0$. Write $F(x) = x \log \frac{1}{x}$ for $x > 0$, which is non-decreasing for $x \le \frac{1}{e}$, and let p_S be the p.m.f. of S on $\mathbb{Z}$. Decomposing the differential entropy over unit intervals,

$$\begin{aligned} h(S + U^{(m)}) &= \int_{\mathbb{R}} F\big(f_{S+U^{(m)}}(x)\big)\, dx \\ &= \sum_{k \in (-5\sigma_S^2,\, 5\sigma_S^2)} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)\, dx + \sum_{|k| \ge 5\sigma_S^2} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)\, dx. \end{aligned} \tag{28}$$

Step 1: the continuous entropy tails. We show that the second term in (28) vanishes as σ_S grows. For $k \ge 5\sigma_S^2$, Proposition 6 of [2] applied to S with $\delta = \frac{1}{2}$ gives $N_{\max} < \sigma_S^2 + 1$ as soon as $\sigma_S > 4$, so that $p_S(k+1) \le p_S(k)$ on that range; moreover $\lfloor x \rfloor \ge N_{\max} + m - 1$ there provided $\sigma_S^2 \ge m/4$, so (22) in Lemma 8 of [2] yields $f_{S+U^{(m)}}(x) \le 2^m p_S(\lfloor x \rfloor - m + 1)$ for $x \in [k, k+1)$. By Lemma 5 of [2] we have $p_{\max} \le 1/\sigma_S$, hence $2^m p_S(\cdot) \le 2^m/\sigma_S \le \frac{1}{e}$ as soon as

$$\sigma_S \;\ge\; 2^m e, \tag{29}$$

which places us in the range where F is non-decreasing. Therefore

$$0 \le \sum_{k \ge 5\sigma_S^2} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)dx \le \sum_{k \ge 5\sigma_S^2} F\big(2^m p_S(k-m+1)\big) \;\le\; 2^{m+4}\, e^{-\sigma_S^{1/5}} \sigma_S^3, \tag{30}$$

where we used Lemma 9 of [2] with $\epsilon = 1/5$ and $\theta = 1-\sigma_S^{-9/5}$, valid provided $\sigma_S \ge \max\{3^5,\, (12e^3)^{5/3}\} = (12e^3)^{5/3}$. The left tail is bounded in exactly the same way using the second part of Lemma 9, giving

$$0 \le \sum_{k \le -5\sigma_S^2} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)dx \le 2^{m+4}\, e^{-\sigma_S^{1/5}} \sigma_S^3. \tag{31}$$

Step 2: the central range is approximately $H(S)$. Writing $M := \sigma_S^2$ and $G(x) := F(x) - x\log M$,

$$\sum_{k \in (-5\sigma_S^2, 5\sigma_S^2)} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)dx = \log(\sigma_S^2)\, \mathbb{P}\Big(S + U^{(m)} \in \big(-\lfloor 5\sigma_S^2 \rfloor, \lfloor 5\sigma_S^2 \rfloor + 1\big)\Big) + \sum_{k \in (-5\sigma_S^2, 5\sigma_S^2)} \int_{[k,k+1)} G\big(f_{S+U^{(m)}}(x)\big)dx.$$

Applying Lemma 12 of [2] with $\mu = \frac{1}{11\sigma_S}$, $D = 2^m\sigma_S$, $M = \sigma_S^2$, $a = f_{S+U^{(m)}}(x)$, $b = p_S(k)$, and using $\sum_k \sup_{x\in[k,k+1)} |g_m(k,x)| \le (2^m - 2)/\sigma_S$ from Lemma 8, the number of terms in the sum being at most $11\sigma_S^2$:

$$\begin{aligned} \sum_{k \in (-5\sigma_S^2, 5\sigma_S^2)} \left| \int_{[k,k+1)} G\big(f_{S+U^{(m)}}(x)\big)dx - G\big(p_S(k)\big) \right| &\le \frac{2\log(11\sigma_S)}{\sigma_S} + \frac{2^{m+1}}{\sigma_S} \log(11e2^m\sigma_S) \\ &\le \frac{3}{4} \cdot \frac{2^{m+2}}{\sigma_S} \log(11e2^m\sigma_S) \\ &\le \frac{2^{m+2}}{\sigma_S} \log(2^{m+2}\sigma_S), \end{aligned} \tag{32}$$

the last inequality using $(11e)^{3/4} \le 4\,\sigma_S^{1/4}$, i.e. $\sigma_S \ge (11e)^3/256 \approx 104.5$.

Consequently,

$$\begin{aligned} &\left| H(S) - \sum_{k \in (-5\sigma_S^2, 5\sigma_S^2)} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)dx \right| \\ &\quad \le \sum_{|k| \ge 5\sigma_S^2} F\big(p_S(k)\big) + \frac{2^{m+2}}{\sigma_S} \log(2^{m+2}\sigma_S) \\ &\qquad + \log(\sigma_S^2) \left| \mathbb{P}\Big(S + U^{(m)} \in \big(-\lfloor 5\sigma_S^2 \rfloor, \lfloor 5\sigma_S^2 \rfloor + 1\big)\Big) - \mathbb{P}\Big(S \in \big(-5\sigma_S^2, 5\sigma_S^2 + 1\big)\Big) \right|. \end{aligned} \tag{33}$$

Step 3: the discrete entropy tails. The computation (30) applies directly with the factor 2^m replaced by 1, since $p_S(k+1) \le \theta p_S(k)$ for $k \ge 5\sigma_S^2$ by Lemma 9. Summing both tails,

$$\sum_{|k| \ge 5\sigma_S^2} F\big(p_S(k)\big) \;\le\; 2^{m+5}\, e^{-\sigma_S^{1/5}} \sigma_S^3. \tag{34}$$

Step 4: the truncation probabilities. Since $\mathbb{E}[S+U^{(m)}] = m/2$ and $\mathrm{Var}(S+U^{(m)}) = \sigma_S^2 + \frac{m}{12}$, and since $5\sigma_S^2 - 1 - \frac{m}{2} \ge 4\sigma_S^2$ whenever $\sigma_S^2 \ge 1 + \frac{m}{2}$, Chebyshev's inequality gives

$$0 \le \mathbb{P}\Big(S + U^{(m)} \notin \big(-\lfloor 5\sigma_S^2 \rfloor, \lfloor 5\sigma_S^2 \rfloor + 1\big)\Big) \le \frac{\sigma_S^2 + m/12}{16\,\sigma_S^4} \le \frac{1}{8\sigma_S^2}, \tag{35}$$

the last step using $\sigma_S^2 \geq m/12$. The same bound holds a fortiori for $\mathbb{P}(S \notin (-5\sigma_S^2, 5\sigma_S^2+1))$. As both probabilities appearing in (33) are also bounded by 1, their difference is at most $\frac{1}{8\sigma_S^2}$, so the third term of (33) is at most $\dfrac{\log \sigma_S^2}{8\sigma_S^2}$.

Step 5: conclusion. Combining (33), (34) and (35),

$$\left| H(S) - \sum_{k \in (-5\sigma_S^2, 5\sigma_S^2)} \int_{[k,k+1)} F\big(f_{S+U^{(m)}}(x)\big)dx \right| \leq 2^{m+5} e^{-\sigma_S^{1/5}} \sigma_S^3 + \frac{2^{m+2}}{\sigma_S} \log\big(2^{m+2}\sigma_S\big) + \frac{\log \sigma_S^2}{8\sigma_S^2},$$

and adding the two continuous tails (30), (31) through (28) yields (16).

Finally we collect the constraints used: $\sigma_S > 4$ (Proposition 6), $\sigma_S^2 \geq m/4$ and $\sigma_S^2 \geq 1 + m/2$ (truncation), $\sigma_S \geq 2^m e$ from (29), $\sigma_S \geq (11e)^3/256$ from (32), and $\sigma_S \geq (12e^3)^{5/3}$ from Lemma 9 with $\epsilon = 1/5$. Since $2^m e < 2^{m+2}$ and $(11e)^3/256 < 3^5 < (12e^3)^{5/3}$, all of these are implied by (15). The right-hand side of (16) tends to 0 as $\sigma_S \to \infty$ for fixed m, which proves the theorem. □

Theorem 5.2. *Let $X_1, \dots, X_n$ be n independent, not necessarily identically distributed, log-concave random variables on $\mathbb{Z}$ with finite variances $\sigma_1^2, \dots, \sigma_n^2$, and set $\sigma_S^2 := \sum_{i=1}^n \sigma_i^2$. Then*

$$H\Big(\sum_i X_i\Big) \geq \frac{1}{n} \sum_i H(X_i) + \frac{1}{2} \log n - o(1),$$

where the $o(1)$-term vanishes as $\sigma_S \to \infty$ for fixed n. More precisely, provided that $\sigma_S \geq \max\{2^{n+2}, (12e^3)^{5/3}\}$,

$$H\Big(\sum_i X_i\Big) \geq \frac{1}{n} \sum_i H(X_i) + \frac{1}{2} \log n - \varepsilon_n(\sigma_S),$$

with ε_n as in (16).

Proof. Let $U_1, \dots, U_n$ be i.i.d. uniforms on $(0,1)$, independent of everything else, and set $\tilde{X}_i := X_i + U_i$, which is absolutely continuous with respect to the Lebesgue measure. By Shannon's entropy power inequality [3] applied to the independent random variables $\tilde{X}_1, \dots, \tilde{X}_n$, together with the arithmetic-geometric mean inequality,

$$N\Big(\sum_i \tilde{X}_i\Big) \;\geq\; \sum_i N(\tilde{X}_i) \;\geq\; n\Big(\prod_i N(\tilde{X}_i)\Big)^{1/n},$$

where $N(X) = \frac{1}{2\pi e} e^{2h(X)}$ denotes the entropy power. Taking logarithms, this is

$$h\Big(\sum_i \tilde{X}_i\Big) \;\geq\; \frac{1}{n} \sum_i h(\tilde{X}_i) + \frac{1}{2} \log n. \tag{36}$$

Write $S := \sum_i X_i$ and $U^{(n)} := \sum_i U_i$, so that $\sum_i \tilde{X}_i = S + U^{(n)}$. By Hoggar's theorem [7], the class of discrete log-concave distributions is closed under convolution, so S is log-concave on $\mathbb{Z}$, with variance $\sigma_S^2 = \sum_i \sigma_i^2$. Applying Theorem 5.1 to S with $m = n$ gives

$$h\big(S + U^{(n)}\big) \;\leq\; H(S) + \varepsilon_n(\sigma_S),$$

which combined with (36) yields

$$H\Big(\sum_i X_i\Big) + \varepsilon_n(\sigma_S) \;\geq\; \frac{1}{n}\sum_i h(\tilde{X}_i) + \frac{1}{2}\log n.$$

Finally, $h(\tilde{X}_i) = H(X_i)$ exactly, with no error term: the density of $\tilde{X}_i = X_i + U_i$ is $f_{\tilde{X}_i}(x) = \sum_{k\in\mathbb{Z}} p_{X_i}(k) f_{U_i}(x-k) = p_{X_i}(\lfloor x \rfloor)$, therefore

$$h(\tilde{X}_i) = -\int_{\mathbb{R}} p_{X_i}(\lfloor x \rfloor)\log p_{X_i}(\lfloor x \rfloor)\,dx = -\sum_{k\in\mathbb{Z}} p_{X_i}(k)\log p_{X_i}(k) = H(X_i).$$

In particular the error in the statement comes only from the left-hand side of (36) and does not accumulate over the n summands. This concludes the proof. □

A.6 Proof of theorem 5.3

Theorem 5.3 (Discrete stability)**.** *Let X_1, X_2 be two independent log-concave random variables on $\mathbb{Z}$, of mean μ_i and variance σ_i^2. We define $Z_i^{(\mathbb{Z})}$ as the Gaussian $\mathcal{N}(\mu_i, \sigma_i^2)$ discretized on $\mathbb{Z}$ and the discrete deficit as:*

$$\delta_H(X_1, X_2) := H(X_1 + X_2) - \frac{1}{2}H(X_1) - \frac{1}{2}H(X_2) - \frac{1}{2}\log 2.$$

Then there exist constants $C_1(\sigma_1, \sigma_2), C_2 > 0$ such that:

$$\frac{D(X_1\|Z_1^{(\mathbb{Z})}) + D(X_2\|Z_2^{(\mathbb{Z})})}{2} \;+\; \frac{1}{2}\log\frac{\sigma_1^2+\sigma_2^2}{2\,\sigma_1\sigma_2} \;\leq\; C_1(\sigma_1,\sigma_2)\,\delta_H(X_1,X_2) \;+\; C_2\,\frac{\log\sigma_{\min}}{\sigma_{\min}}.$$

Proof. Let U_1, U_2 be two iid random variables of distribution $\mathcal{U}(0,1)$ on $\mathbb{R}$. We first define $\tilde{X}_i := X_i + U_i$ which is real valued and absolutely continuous wih respect to the Lebesgue measure. We then let $\tilde{\sigma}_i^2 := Var(\tilde{X}_i) = \sigma_i^2 + \frac{1}{12}$ as well as $s^2 := \frac{1}{2}(\tilde{\sigma}_1^2 + \tilde{\sigma}_2^2)$ and $V_i := \tilde{X}_i/s$ so that $\frac{1}{2}(Var(V_1) + Var(V_2)) = 1$.

Step 1:

By Proposition 12 of [1], applied separately to X_1 and to X_2, we have $C_P(\widetilde{X}_i) \leq 438244\,\widetilde{\sigma}_i^2$. Since the Poincare constant is 2-homogeneous, $C_P(aX) = a^2 C_P(X)$, and since $\widetilde{\sigma}_i^2 < \tilde{\sigma}_1^2 + \tilde{\sigma}_2^2 = 2s^2$, we obtain the absolute bound

$$R^*_{V_i} \;=\; \frac{C_P(\widetilde{X}_i)}{s^2} \;\leq\; 438244\,\frac{\widetilde{\sigma}_i^{\,2}}{s^2} \;<\; 876\,488 \;=:\; R,$$

which does not depend on the scale of the X_i. As $R^*_G = 1 \leq R$ for a standard Gaussian G, the hypothesis $\max\{R^*_{V_1}, R^*_{V_2}, R^*_G\} \leq R$ of Theorem 3.4 of [10] is satisfied.

Moreover, writing $\rho := \widetilde{\sigma}_2/\widetilde{\sigma}_1 \geq 1$,

$$a \;:=\; \min_i \mathrm{Var}(V_i) \;=\; \frac{\widetilde{\sigma}_1^{\,2}}{s^2} \;=\; \frac{2\widetilde{\sigma}_1^{\,2}}{\widetilde{\sigma}_1^{\,2} + \widetilde{\sigma}_2^{\,2}} \;=\; \frac{2}{1+\rho^2} \;\leq\; 1,$$

so that $\min\{a, 1\} = a$. Adding $\frac{1}{12}$ to both variances brings their ratio closer to 1, therefore $\rho \leq \sigma_2/\sigma_1$. Theorem 3.4 of [10] applied to (V_1, V_2) thus yields

$$\frac{1}{C_1(\sigma_1,\sigma_2)}\left(\frac{1}{2}\log(2\pi e) - \frac{h(V_1)+h(V_2)}{2}\right) \;\leq\; \delta_{EPI,1/2}(V_1,V_2), \qquad C_1(\sigma_1,\sigma_2) := 1 + \frac{2R}{a}. \quad (37)$$

Step 2:

As $D(\tilde{X}_i) = \frac{1}{2}\log(2\pi e\tilde{\sigma}_i^2) - h(\tilde{X}_i)$ and $h(V_i) = h(\tilde{X}_i) - \log s$, we notice that:

$$\frac{1}{2}\log(2\pi e) - \frac{h(V_1)+h(V_2)}{2} = \frac{1}{2}\log(2\pi e s^2) - \frac{h(\tilde{X}_1)+h(\tilde{X}_2)}{2}$$

And

$$\frac{1}{2}\log(2\pi e s^2) = \frac{1}{2}\log(2\pi e\frac{\tilde{\sigma}_1\tilde{\sigma}_2 s^2}{\tilde{\sigma}_1\tilde{\sigma}_2}) = \frac{1}{2}\left(\frac{1}{2}\log(2\pi e\tilde{\sigma}_1^2) + \frac{1}{2}\log(2\pi e\tilde{\sigma}_2^2)\right) + \frac{1}{2}\log(\frac{s^2}{\tilde{\sigma}_1\tilde{\sigma}_2})$$

We therefore have:

$$\frac{1}{2}\log(2\pi e) - \frac{h(V_1)+h(V_2)}{2} = \frac{D(\tilde{X}_1)+D(\tilde{X}_2)}{2} + \frac{1}{2}\log(\frac{s^2}{\tilde{\sigma}_1\tilde{\sigma}_2})$$

Then, using $h(\tilde{X}_i) = H(X_i)$ and using the scale invariance of δ_{EPI}, we deduce:

$$\delta_{EPI,1/2}(V_1,V_2) = \delta_{EPI,1/2}(\tilde{X}_1,\tilde{X}_2) = h(S+U^{(2)}) - \frac{1}{2}\log 2 - \frac{H(X_1)+H(X_2)}{2}$$

With $S = X_1 + X_2$ and $U^{(2)} = U_1 + U_2$. Using Hoggar's result, S is log-concave of variance $\sigma_1^2+\sigma_2^2$ and using Theorem 5.1 with $m=2$, we have $h(S+U^{(2)}) \leq H(S) + \epsilon(\sigma_S)$

We deduce $\delta_{EPI,1/2}(V_1,V_2) \leq \delta_H(X_1,X_2) + \epsilon(\sigma_S)$

Step 3:

As the maximal span is equal to 1 with the log-concavity, using the theorem 2.1 in [6] with $n=1$, we get:

$$D(X_i\|Z_i^{(\mathbb{Z})}) \leq \frac{1}{2}\log(2\pi e\sigma_i^2) - H(X_i) + \frac{2}{\sigma_i} \leq \frac{1}{2}\log(2\pi e\tilde{\sigma}_i^2) - h(\tilde{X}_i) + \frac{2}{\sigma_i} = D(\tilde{X}_i) + \frac{2}{\sigma_i}$$

Putting everything together, we get:

$$\frac{D(X_1\|Z_1^{(\mathbb{Z})}) + D(X_2\|Z_2^{(\mathbb{Z})})}{2} + \frac{1}{2}\log\frac{\sigma_1^2+\sigma_2^2+1/6}{2\sqrt{(\sigma_1^2+1/12)(\sigma_2^2+1/12)}} \leq C_1(\sigma_1,\sigma_2)[\delta_H + \epsilon(\sigma_1^2+\sigma_2^2)] + \frac{1}{\sigma_1} + \frac{1}{\sigma_2} \tag{38}$$

□

References

[1] L. Gavalakis and I. Kontoyiannis, Conditions for equality and stability in Shannon's and Tao's entropy power inequalities, preprint arXiv:2509.14021, 2025.

[2] L. Gavalakis, Approximate discrete entropy monotonicity for log-concave sums, Combinatorics, Probability and Computing, vol. 33, no. 2, pp. 196–209, 2024.

[3] A. J. Stam, Some inequalities satisfied by the quantities of information of Fisher and Shannon, Information and Control, vol. 2, no. 2, pp. 101–112, 1959.

[4] S. G. Bobkov and G. P. Chistyakov, Entropy power inequality for the Rényi entropy, IEEE Transactions on Information Theory, vol. 61, no. 2, pp. 708–714, Feb. 2015.

[5] A. R. Barron, Monotonic Central Limit Theorem for Densities, Technical Report No. 50, Department of Statistics, Stanford University, March 1984.

[6] L. Gavalakis and I. Kontoyiannis, Entropy and the discrete central limit theorem, Stoch. Proc. Appl., 170:104294, June 2024.

[7] S. G. Hoggar, Chromatic polynomials and logarithmic concavity, Journal of Combinatorial Theory, Series B, vol. 16, no. 3, pp. 248–254, 1974.

[8] C. E. Shannon, A mathematical theory of communication, Bell System Technical Journal, vol. 27, no. 3, pp. 379–423, 623–656, 1948.

[9] A. R. Barron, Entropy and the Central Limit Theorem, The Annals of Probability, vol. 14, no. 1, pp. 336–342, 1986.

[10] L. Yao and S. Yuan, Entropy Jump and Entropic Central Limit Theorem for Independent Sum, preprint arXiv:2402.08953, 2024.

[11] P. Billingsley, Convergence of Probability Measures, John Wiley & Sons, New York, NY, second edition, 1999.

[12] E. H. Lieb, Proof of an entropy conjecture of Wehrl, Communications in Mathematical Physics, vol. 62, no. 1, pp. 35–41, 1978.

[13] N. M. Blachman, The convolution inequality for entropy powers, *IEEE Trans. Inform. Theory*, vol. IT-11, no. 2, pp. 267–271, 1965.

[14] E. A. Carlen and A. Soffer, Entropy production by block variable summation and central limit theorems, *Comm. Math. Phys.*, vol. 140, no. 2, pp. 339–371, 1991.

[15] M. H. M. Costa and T. M. Cover, On the similarity of the entropy power inequality and the Brunn-Minkowski inequality, *IEEE Trans. Inform. Theory*, vol. IT-30, no. 6, pp. 837–839, 1984.

[16] A. Dembo, T. M. Cover, and J. A. Thomas, Information theoretic inequalities, *IEEE Trans. Inform. Theory*, vol. 37, no. 6, pp. 1501–1518, 1991.

[17] M. Fradelizi, L. Gavalakis, and M. Rapaport, "On the monotonicity of discrete entropy for log-concave random vectors on $\mathbb{Z}^d$," *Discrete & Computational Geometry*, to appear. Preprint arXiv:2401.15462, 2024.

[18] T. Tao, *Sumset and inverse sumset theory for Shannon entropy*, Combinatorics, Probability and Computing, vol. 19, no. 4, pp. 603–639, 2010.